

New component innovations in 2015

Lower capacitance to avoid interfering with high-speed data transfer. ESD protection devices must offer circuit protection without interfering with the daily functionality of the circuit being protected. For example, on an RF interface (like Bluetooth or ZigBee) or a wired port (like USB 2.0), the ESD protector must not cause distortion or loss of strength of data signals. To provide signal integrity, the capacitance of the ESD protector must be minimised without compromising protection levels. New transient voltage suppression (TVS) diodes feature a capacitance value of 0.35pF to ensure that these will remain invisible to high-speed signals.

Lower clamping voltage to protect even the most sensitive circuits. If an ESD event occurs, the primary job of the ESD protector is to divert and dissipate as much of the ESD transient as possible. This characteristic is improved by reducing the on-state resistance or dynamic resistance. By decreasing the dynamic resistance, the ESD protector carries significantly more of the surge current than the circuit being protected. By doing this, it reduces the electrical stress on the integrated circuit (IC) and ensures its survival.

Smaller form factors to fit the limited board space available in wearable devices. No matter how well a protection device performs, it is not very useful if it cannot fit into the application it is meant to protect. As wearable devices get thinner and smaller, circuit boards will have minimal space available to accommodate ESD protection solutions. Discrete diodes are the best solution to this potential design challenge because these give design engineers exceptional board-layout flexibility.

Key features of next-gen devices

- Use of sensor-fusion chips to reduce device size and software development time
- A smartphone converts media into lightweight version to reduce overhead on the wearable device
- Bluetooth Low Energy (BLE) is becoming the most popular means of communication
- Work with sensor data ranges rather than absolute values to reduce reliance on multiple sensors
- Increased specialisation of devices, resulting in removal of unnecessary sensors
- ESD protection that is invisible to high-speed signals
- Extremely-high-resolution displays with very wide viewing angles and a better colour gamut

Data processing needs to be moved to the gateway; in the case of most wearables, this gateway is the smartphone

an interesting challenge to circuit designers. Why? Think about how these devices are designed to get close and personal with the consumer. Because these are meant to be worn on the skin, there is significant risk of exposure to user-generated static electricity. For instance, Jawbone wristband uses a Littelfuse polymeric positive temperature coefficient (PPTC) device resettable fuse.

Unfortunately, simple human touch can generate transient electrostatic discharge (ESD). Without proper protection, any sensor circuits, battery-charging interfaces, buttons

or data I/Os could provide a path for ESD to enter the wearable device and cause irreparable damage.

James Colby, manager, business and technology development, semiconductor business unit, Littelfuse, says, "Circuit designers for wearables are tasked with incorporating advanced circuit-protection technologies that safeguard the device as well as the user from irreparable damage."

Colby adds, "Another challenge faced by circuit designers stems from the shrinking form factor of most wearables and the need to accomplish improved performance with a smaller estate."

Design displays with ambient light in mind

The scene of people bending their necks to peer into their phones is common. It is easy to hold the phone such that the screen is parallel to the

face, but is it really that easy to hold it like that, while, say, running to catch a bus?

Smartwatches will, more often than not, be used at less than optimal angles. If the display used in the device has a limited viewing angle, it will stress the eyes and, eventually, frustrate the user.

Even if the viewing angle is perfect, ambient light needs to be considered for these devices. The colour gamut of the watch screen will need to be much better as there is significant washout due to ambient light.

If the display is not bright enough or lacks the contrast to be viewed in direct sunlight, that is going to be another problem. Liquid crystal display (LCD) and organic light emitting diode (OLED) technologies are the ones mostly used in smartwatches; Samsung Galaxy Gear 2 comes with a super-AMOLED RGB Stripe display.

Keeping in tune with the saying, "If you cannot beat them, join them," watches like Sony SmartWatch 2 and Pebble use ambient light to their advantage instead of fighting it. This is done by using transreflective LCD technology with a low-pixel aspect ratio (ratio of pixel width to height). The transreflective technology reflects some of the ambient light hitting it.

Resolution is another important element. With most users already accustomed to high resolution on their smartphones, they expect a similar pixilation-free experience on their smartwatches. Of course, simple or more specialised devices might not require a screen at all; these simply make do with a mix of blinking lights or haptic feedback.

Design element that matters

While researching for this article, if there is one thing that stood out, it was the answer to the question: What is the most important design characteristic in a device?

The unanimous response was that the device should look fashionable or cool, be intuitive and very comfortable to use. This could be the